

HCV Module for Viessmann Vitocal Heat Pumps

Description

HCV (Home Control Viessmann) is an adapter that integrates Viessmann heat pumps supporting E3 communication via CAN into a home control environment, without requiring advanced technical knowledge. In other words, apart from connecting the HCV module to the internal CAN bus and configuring Wi-Fi access, no further settings or expertise are required.

Vitocal 250	
DomesticHotWaterSensor	51.2
OutsideTemperatureSensor	17.8
MixerOneCircuitFlowTemperatur	25.5
WaterPressureSensor	1.7
PrimaryHeatExchangerLiquidTem	3.9
CentralHeatingPump	on
DomesticHotWaterPump	on
MixerOneCircuitPump	on
MixerOneCircuitRoomTemperatur	23.0
Time	16:48
EnergyConsumptionCentralHeati	53.6
MixerOneCircuitOperationState	on
PrimaryInletTemperatureSensor	17.7
SecondaryOutletTemperatureSen	26.8
PrimaryCircuitFanOne	32
PrimaryCircuitFanTwo	37
EvaporatorVaporTemperatureSen	7.9
CompressorSpeedPercent	20
HeatPumpCompressor	on
CurrentThermalCapacitySystem	3669

Once installed, data retrieved via the CAN bus can be viewed using any local web browser. If integration into a home control environment is not required, the installation process is complete at this stage. Current data can then be queried at any time by navigating to the address **hcv.local** in a browser.

Initial Installation and Reset

By default, the integrated Access Point mode is active, allowing the HCV module to be accessed via the address `192.168.1.1`. This state is indicated by a flashing red LED.



The network named "HCV" will appear in your mobile phone's network settings and can be selected. Once connected, you can enter and save your local network's credentials (SSID and password). To switch to standard Wi-Fi mode: disconnect the power supply (USB plug), flip switch 2 to the "OFF" position, and reconnect the power. Once the HCV module's status LED flashes green, the device can be accessed via the regular Wi-Fi network using the address `hcv.local`.

Note: Under the "Vitocal Data" menu item—which also serves as the HCV module's main page—the message "No device detected" is displayed if no valid CAN messages are detected. The display updates once at least one valid message is recognized. The messages in the "Vitocal 250" window are displayed in either German or the origin defined E3onCan messages.

In the event of an issue, the local network settings can be modified at any time by flipping switch 2 to the "ON" position and subsequently restarting the device.

MQTT Broker Configuration

The screenshot shows a web interface titled "Home Control Viessmann V:0.83". Below the title bar is a section titled "MQTT Konfiguration". It contains four input fields: "Broker IP" with the value "111.222.33.4", "Broker Port" with the value "1883", "Broker User" with the value "User", and "Broker Password" with masked characters "*****". At the bottom of this section is a blue button labeled "speichern".

Once a connection has been successfully established, the broker's address—and, if applicable, the username and password—can be entered via the "MQTT Broker Config" menu option. Saving the configuration data triggers a restart of the HCV module.

As soon as CAN messages valid for a Vitocal heat pump are detected, they also appear as MQTT messages on the local network or within the MQTT broker.

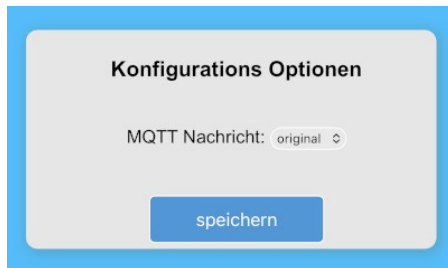
Note: The values (DIDs) currently defined in the Open3E datapoints are the ones transmitted.

We would like to take this opportunity to thank the creators of the document that served as the basis for the CAN messages for their work.

As previously mentioned, the CAN messages displayed in the browser are translated into German. The same setting can be applied to MQTT messages by selecting "German" under the "HCV Options" menu item. The advantage of the German version is that the MQTT messages are shorter.

MQTT Broker Message Format

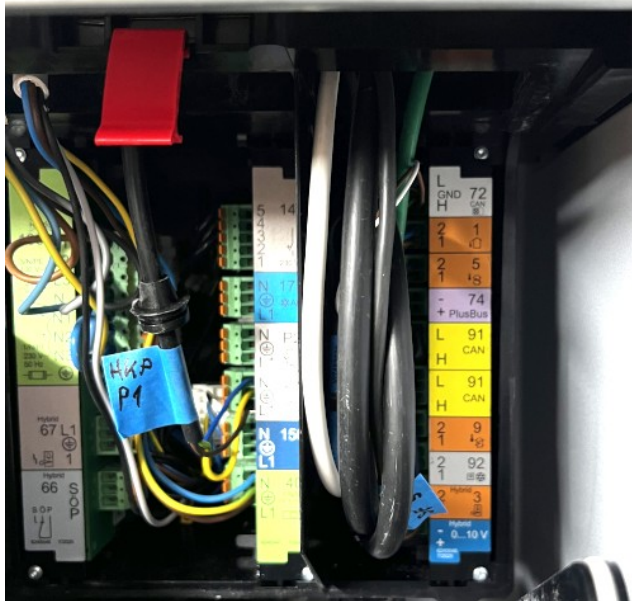
The MQTT messages displayed and sent by the HCV module can be customized.

The screenshot shows a web interface titled "Konfigurations Optionen". It contains a single dropdown menu labeled "MQTT Nachricht:" with the value "original" selected. At the bottom of this section is a blue button labeled "speichern".

Either the data IDs defined in the OpenE3datapoints document are used, or the German-language text strings are employed. This selection can be found under the "HCV Options" menu item.

CAN Bus Installation

The connection for detecting messages from Viessmann heat pumps is made via the internal CAN bus connector 72 of the Vitocal indoor unit. Since this connection is



already in use, the HCV module's sensor cable must be connected in parallel. Pay attention to the CAN bus polarity when doing so. The terminal on the HCV module corresponding to CAN High is marked in black.

Setting switch 1 to the "ON" position activates the CAN terminating resistor. This must be switched off when using the internal connection!

Note: The maximum length of the cable between the Vitocal connection and the HCV module connector must not exceed **10 meters**.

It should also be noted that the CAN connection is galvanically isolated to

prevent interference on both the CAN bus and the connected devices.

Status LED

LED flashes red: Access Point mode is selected via switch 2, or there is no connection to the configured local Wi-Fi.

LED flashes red and green: Connection to the local Wi-Fi is established, but there is no MQTT connection to a broker.

LED flashes green: Wi-Fi and MQTT connections are active.

HCV Firmware Update

The HCV module's software can be updated via OTA (Over-the-Air), meaning directly over the network. Current versions are available on the embdes.com website and under [github /chrkir/hcv](https://github.com/chrkir/hcv). In the HCV menu, you will find this function under "Firmware update".



Support

For questions, errors, or suggestions, please write to info@embdes.com.

Configuration Settings

The main page displays configuration data read from the `vcalobj.did` file. The corresponding values are updated based on received CAN messages and simultaneously transmitted via the MQTT connection. This page can be accessed via the "Vitocal Data" menu option.



The "HCV Info" menu option displays function-relevant data as well as the internal memory allocation.

The "Settings" option is used to configure CAN communication settings. The control software you are using is a key factor here. A JSON format has been implemented specifically for home automation software, allowing for the automatic registration of data to be transmitted. All other systems operate using the "MQTT Individual Messages" selection or, in some cases, the JSON format. For all variants, it is important not to change the selected MQTT format after initial testing; otherwise, the application software may fail to process the messages.

Wi-Fi configuration allows you to set the SSID and password.

The MQTT Broker configuration section is used to set the broker IP, port, username, and password.

"Firmware Update" allows you to install a new software version; these versions can be downloaded from embdes.com or [github/chrkir/hcv](https://github.com/chrkir/hcv).

"Import Vitocal Objects" loads either the basic configuration provided by embdes or an extended/modified variant into the HCV adapter. The settings for the data to be imported are described in the `vcalobj.did` file. If in doubt, please contact embdes for support.

The "Import Software License" function allows you to activate self-built hardware for use as licensed hardware. Hardware supplied by embdes is already activated.

Messages regarding events relevant to the adapter's operation are written to the log file. These messages can be viewed, transmitted, and deleted using the corresponding function. CAN Log Data Download: If CAN logging is enabled in the settings, this option allows you to download the stored CAN log data to your PC. Before the transfer, the data is converted into a Linux-compatible candump format. This data can be used to extend or customize the Viessmann Data IDs (DIDs) for analysis purposes. Relevant information can be found at: [github/..e3oncan](https://github.com/chrkir/hcv).

HCV MQTT ioBroker - Home Control – Node red

If the MQTT messages are to be integrated into a home automation application, this can be done by simply selecting the corresponding objects in the ioBroker MQTT tree. Other home control applications also accept this format; in some cases, the MQTT variant in JSON format is supported as well, offering the advantage of reduced system load.

File Explorer	Model Explorer	Model Explorer	Model Explorer
mqtt			
0			
1054	1054	folder	
EVC	EVC	state	variable
HCV	HCV	folder	
vcal	vcal	folder	
AllengraSensor	HCV/vcal/AllengraSensor	state	variable
BackendConnectionStatus	HCV/vcal/BackendConnect	state	variable
CentralHeatingPump	HCV/vcal/CentralHeatingP	state	variable
CompressorInletPressureSensor	HCV/vcal/CompressorInlet	state	variable
CompressorInletTemperatureSensor	HCV/vcal/CompressorInlet	state	variable
CompressorOilTemperatureSensor	HCV/vcal/CompressorOilTe	state	variable
CompressorOutletPressureSensor	HCV/vcal/CompressorOutle	state	variable
CompressorOutletTemperatureSensor	HCV/vcal/CompressorOutle	state	variable
CompressorSpeedPercent	HCV/vcal/CompressorSpec	state	variable
CurrentThermalCapacitySystem	HCV/vcal/CurrentThermalC	state	variable
DomesticHotWaterPump	HCV/vcal/DomesticHotWat	state	variable
DomesticHotWaterSensor	HCV/vcal/DomesticHotWat	state	variable
EconomizerLiquidTemperatureSensor	HCV/vcal/EconomizerLiqui	state	variable

HCV MQTT Home Assistant

As previously mentioned in the "Message Format" section, the HCV supports Home Assistant via a custom MQTT implementation. If the MQTT service is installed and enabled in Home Assistant, the Viessmann Vitocal heat pump is automatically detected and installed. Once the HCV module is connected to the internal CAN bus, automatic registration is initiated upon powering up the module. The values received from the control unit are then displayed in the Home Assistant "Devices" overview and continuously updated.

Viessmann Heat Pump >		Bereich zuordnen
	DomesticHotWaterSensor	51,2 °C
	OutsideTemperatureSensor	17,8 °C
	FlowTemperatureSensor	27,1 °C
	ReturnTemperatureSensor	25,1 °C
	HydraulicSeparatorTemperatureSensor	25,5 °C
	MixerOneCircuitFlowTemperatureSensor	25,5 °C
	WaterPressureSensor	1,70 bar
	PrimaryHeatExchangerLiquidTemperatureSensor	3,9 °C
	CompressorInletTemperatureSensor	6,4 °C
	CompressorInletPressureSensor	4,20 bar
	CompressorOutletTemperatureSensor	37,8 °C
	CompressorOutletPressureSensor	11,90 bar
	SecondaryHeatExchangerLiquidTemperatureSensor	25,3 °C
	CentralHeatingPump	In Betrieb
	DomesticHotWaterPump	Außer Betrieb

Firmware Changes:

Version: 0.89

Support for Home Assistant.

Version: 0.90

Configuration file upload for Data IDs (DIDs) via the Settings menu; MQTT password length increased to 32 characters. Protocol control via a dedicated display.

Version: 0.91

CAN logging added. Messages not already loaded via the `vcalobj.did` configuration settings are recorded in Linux `candump` format.

Version: 1.01

Active control of the Vitocal (write operations for DIDs):

Functionality has been expanded to include writing Data IDs (DIDs). Specific Vitocal parameters can now be adjusted via an MQTT message in JSON format.

- Transmit topic: `HCV/writeDid`
- Example payload (set room temperature setpoint for mixer circuit 1 to 20 °C):

```
`{"did": 424, "value": 200}`
```

(Note: The value 200 corresponds to 20.0 °C, as this data point is defined as `Float\_10` in `vcalobj.did` and is scaled by a factor of 10.)

- Feedback / Status:

The response from the Vitocal controller is reported back transparently via MQTT:

- Topic: `HCV/writeDidResponse`
- Payload: `{"status": "success"}` or `{"status": "error"}`
- Filter function for CAN logging:

CAN logging now offers a flexible filter function. Users can choose whether to record all messages on the CAN bus or only specific CAN IDs. This significantly simplifies analysis when extending pre-configured MQTT messages or identifying new data points.

- Recorded data can be downloaded to the PC's local download folder via the "CAN Log Data Download" menu item. Since this data is available in a binary format, it can be converted into a readable text format using the integrated CAN Log Converter tool. This

format can also be played back for simulation purposes using the Linux program ``canplayer``.

- Enhanced DID handling & configuration updates:

The new data structure handling allows for more flexible management and updating of DIDs, enabling the integration of new control and read options into the system with minimal effort.

- Optimized user interface:

Improved display of help text and parameter descriptions in the web interface, ensuring a clearer distinction between setting explanations and live measurement values.

Data objects that can be modified are highlighted in color (blue).